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PENJELASAN PROGRAM PENYAKIT THT - PERTEMUAN 5

SOURCE CODE

1. Import Library

```
import tkinter as tk
from tkinter import messagebox
```

tkinter: membuat tampilan GUI
messagebox: menampilkan popup hasil diagnosa

2. Data Gejala

```
# DATA GEJALA
gejala = {
    "G1": "Nafas abnormal",
    "G2": "Suara serak",
    "G3": "Perubahan kulit",
    "G4": "Telinga penuh",
    "G5": "Nyeri bicara menelan",
    "G6": "Nyeri tenggorokan",
    "G7": "Nyeri leher",
    "G8": "Pendarahan hidung",
    "G9": "Telinga berdenging",
    "G10": "Airliur menetes",
    "G11": "Perubahan suara",
    "G12": "Sakit kepala",
    "G13": "Nyeri pinggir hidung",
    "G14": "Serangan vertigo",
    "G15": "Getah bening",
    "G16": "Leher bengkak",
    "G17": "Hidung tersumbat",
    "G18": "Infeksi sinus",
    "G19": "Beratbadan turun",
    "G20": "Nyeri telinga",
    "G21": "Selaput lendir merah",
    "G22": "Benjolan leher",
    "G23": "Tubuh tak seimbang",
    "G24": "Bolamata bergerak",
    "G25": "Nyeri wajah",
    "G26": "Dahi sakit",
```

```
    "G27": "Batuk",
    "G28": "Tumbuh dimulut",
    "G29": "Benjolan dileher",
    "G30": "Nyeri antara mata",
    "G31": "Radang gendang telinga",
    "G32": "Tenggorokan gatal",
    "G33": "Hidung meler",
    "G34": "Tuli",
    "G35": "Mual muntah",
    "G36": "Letih lesu",
    "G37": "Demam"
}
```

Menyimpan kode gejala (G1–G37) dan deskripsinya, digunakan untuk di tampilkan ke user sebagai pilihan. Ini disebut fakta (Fact Base) dalam system pakar.

3. Rule Penyakit

```
# RULE PENYAKIT
penyakit = {
    "Tonsilitis": ["G37","G12","G5","G27","G6","G21"],
    "Sinusitis Maksilaris": ["G37","G12","G27","G17","G33","G36","G29"],
    "Sinusitis Frontalis": ["G37","G12","G27","G17","G33","G36","G21","G26"],
    "Sinusitis Edmoidalis":
["G37","G12","G27","G17","G33","G36","G21","G30","G13","G26"],
    "Sinusitis Sfenoidalis": ["G37","G12","G27","G17","G33","G36","G29","G7"],
    "Abses Peritonsiler": ["G37","G12","G6","G15","G2","G29","G10"],
    "Faringitis": ["G37","G5","G6","G7","G15"],
    "Kanker Laring": ["G5","G27","G6","G15","G2","G19","G1"],
    "Deviasi Septum": ["G37","G17","G20","G8","G18","G25"],
    "Laringitis": ["G37","G5","G15","G16","G32"],
    "Kanker Leher & Kepala": ["G5","G22","G8","G28","G3","G11"],
    "Otitis Media Akut": ["G37","G20","G35","G31"],
    "Contact Ulcers": ["G5","G2"],
    "Abses Parafaringeal": ["G5","G16"],
    "Barotitis Media": ["G12","G20"],
    "Kanker Nafasoring": ["G17","G8"],
    "Kanker Tonsil": ["G6","G29"],
    "Neuronitis Vestibularis": ["G35","G24"],
    "Meniere": ["G20","G35","G14","G4"],
    "Tumor Syaraf Pendengaran": ["G12","G34","G23"],
    "Kanker Leher Metastatik": ["G29"],
    "Osteosklerosis": ["G34","G9"],
    "Vertigo Postular": ["G24"]
}
```

Menyimpan hubungan antara gejala dan penyakit.

4. INISIALISASI GUI

```
# INISIALISASI GUI
root = tk.Tk()
root.title("Sistem Pakar Diagnosa Penyakit THT")
root.geometry("620x600")
```

Membuat jendela utama aplikasi dan mengatur judul juga ukuran window.

5. Judul Aplikasi

```
# JUDUL APLIKASI
tk.Label(root, text="SISTEM PAKAR DIAGNOSA PENYAKIT THT",
         font=("Arial", 14, "bold")).pack(pady=10)
```

Menampilkan teks judul di bagian atas GUI.

6. Frame & Scroll

```
# FRAMEE SCROLL
frame_canvas = tk.Frame(root)
frame_canvas.pack(fill="both", expand=True)

canvas = tk.Canvas(frame_canvas)
scrollbar = tk.Scrollbar(frame_canvas, orient="vertical", command=canvas.yview)
scrollable_frame = tk.Frame(canvas)
```

Menampung banyak checkbox dan menambah scroll agar bias digeser (karena banyak gejala).

7. Konfigurasi Scroll

```
#KONFIGURASI SCROLL
scrollable_frame.bind(
    "<Configure>",
    lambda e: canvas.configure(scrollregion=canvas.bbox("all"))
)

canvas.create_window((0, 0), window=scrollable_frame, anchor="nw")
canvas.configure(yscrollcommand=scrollbar.set)

canvas.pack(side="left", fill="both", expand=True)
scrollbar.pack(side="right", fill="y")
```

Mengatur agar scroll mengikuti jumlah konten dan supaya semua gejala bisa terlihat.

8. Checkbox Gejala

```
#CHECKBOX GEJALA
check_vars = {}

tk.Label(scrollable_frame, text="Pilih Gejala:",
         font=("Arial", 12, "bold")).pack(anchor="w")

for kode, nama in gejala.items():
    var = tk.IntVar()
    tk.Checkbutton(scrollable_frame, text=f"{kode} - {nama}",
                   variable=var).pack(anchor="w")
    check_vars[kode] = var
```

Menampilkan semua gejala dalam bentuk checkbox , sehingga user bisa memilih gejala yang dialami.

Check_vars : itu untuk menyimpan status centang (1) atau tidak (0).

9. Fungsi Diagnosa

```
# FUNGSI DIIAGNOSA
def diagnosa():
    dipilih = [k for k, v in check_vars.items() if v.get() == 1]

    # Validasi minimal gejala
    if len(dipilih) < 2:
        messagebox.showwarning("Peringatan", "Pilih minimal 2 gejala!")
        return

    hasil = {}
    for nama_penyakit, rule in penyakit.items():
        cocok = sum(1 for g in rule if g in dipilih)
        hasil[nama_penyakit] = cocok / len(rule)

    terbaik = max(hasil, key=hasil.get)
    nilai_terbaik = hasil[terbaik]

    # Validasi minimal kecocokan
    if nilai_terbaik < 0.5:
        messagebox.showinfo("Hasil Diagnosa", "Gejala tidak cukup untuk
menentukan penyakit.")
```

```
        return

    messagebox.showinfo(
        "Hasil Diagnosa",
        f"Penyakit: {terbaik}\nTingkat Kecocokan: {int(nilai_terbaik*100)}%"
    )
```

Fungsi diagnosa digunakan untuk mencocokkan gejala yang dipilih user dengan data penyakit, kemudian menghitung tingkat kecocokan dan menampilkan penyakit dengan nilai tertinggi.

10. Fungsi Riset

```
# FUGSI RESET
def reset():
    for var in check_vars.values():
        var.set(0)
frame_tombol = tk.Frame(root)
frame_tombol.pack(pady=10)
```

Menghapus semua centang checkbox dan mengembalikan ke kondisi awal.

11. Tombol

```
# FRAME TOMBOL
tk.Button(frame_tombol, text="Reset", bg="red", fg="white",
          command=reset).pack(side="left", padx=10)

tk.Button(frame_tombol, text="Diagnosa", bg="green", fg="white",
          command=diagnosa).pack(side="left", padx=10)
```

Reset: menghapus input

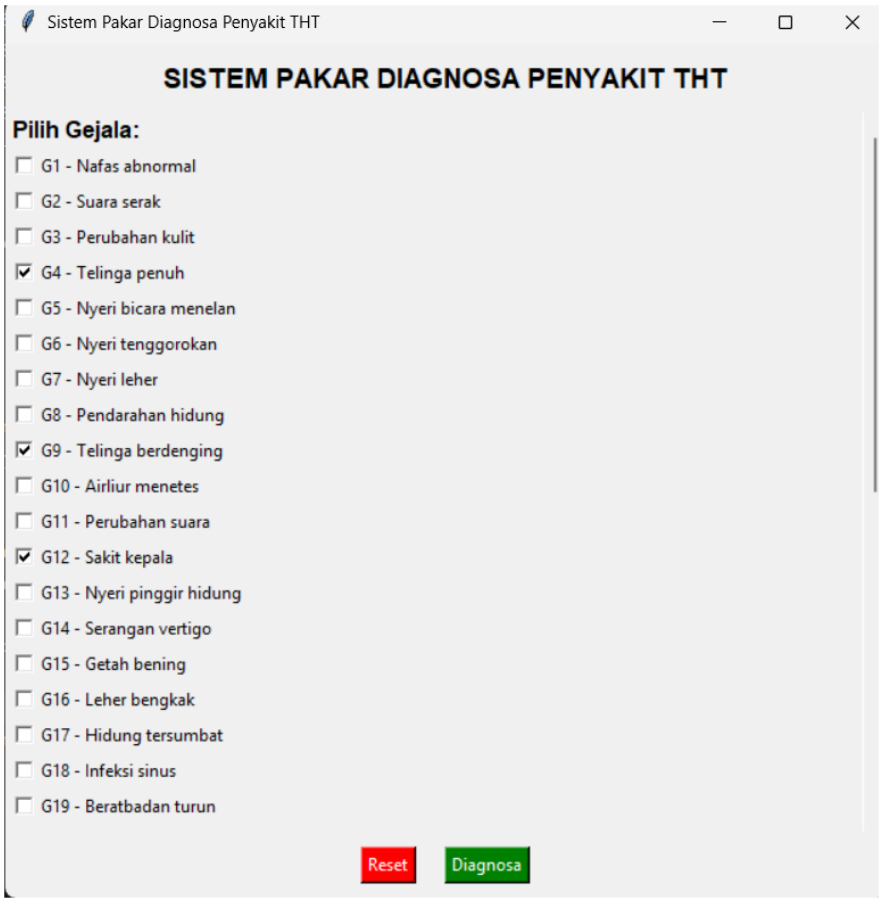
Diagnosa: menjalankan proses diagnosa

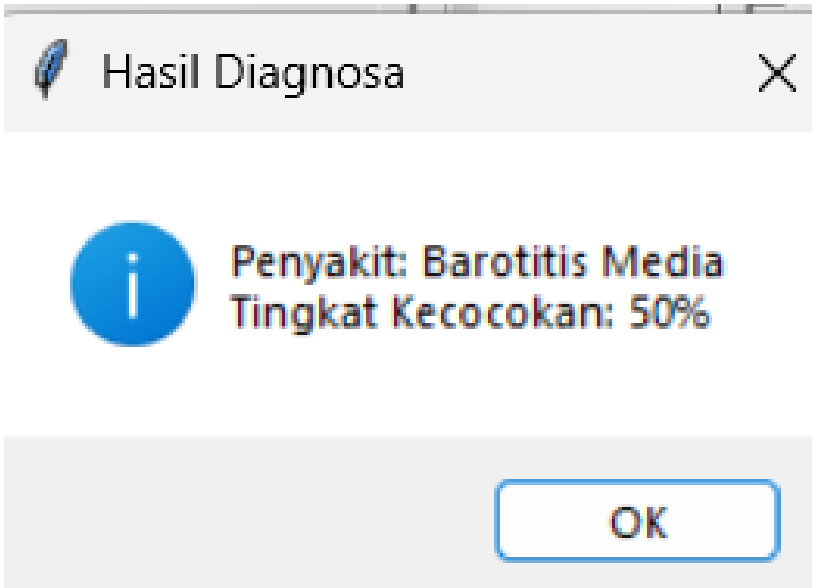
12. Mainloop

```
# JALANKAN APLIKASI
root.mainloop()
```

Menjalankan aplikasi dan menunggu interaksi user.

OUTPUT





Output dari sistem berupa hasil diagnosa penyakit THT berdasarkan gejala yang dipilih oleh user. Sistem menampilkan nama penyakit dengan tingkat kecocokan tertinggi dalam bentuk persentase.